

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 3 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Time Response of Second-Order Systems

A **second-order system** is the most important standard form in control engineering. Its response to a unit step input exhibits all the key transient characteristics — rise, overshoot, oscillation, and settling — that are used to specify and evaluate system performance.

Standard Second-Order Transfer Function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where ω_n = **natural frequency** (rad/s) and ζ = **damping ratio** (dimensionless).
The **characteristic equation** is $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$.

1.1 Poles of the Second-Order System

The two closed-loop poles are:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

For $0 < \zeta < 1$ (underdamped): $s_{1,2} = -\zeta\omega_n \pm j\omega_d$, where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ is the **damped natural frequency**.

1.2 Damping Cases

Condition	Pole Location	Response
$\zeta = 0$	$\pm j\omega_n$ (imaginary)	Undamped — sustained sinusoidal oscillation
$0 < \zeta < 1$	Complex conjugate in LHP	Underdamped — oscillatory, decaying; <i>most common in practice</i>
$\zeta = 1$	Two equal real poles	Critically damped — fastest non-oscillatory response
$\zeta > 1$	Two distinct real poles	Overdamped — sluggish, no oscillation

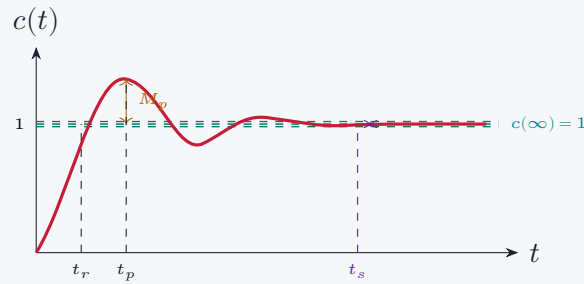
1.3 Unit Step Response (Underdamped, $\zeta < 1$)

Step Response Formula

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t + \phi), \quad t \geq 0$$

where $\phi = \cos^{-1}(\zeta)$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$.

Typical Underdamped Step Response



2. Performance Specifications in Time Domain

Time-domain specifications quantify the transient and steady-state behaviour of a system's step response. They are used to set design requirements and evaluate controller performance.

2.1 Transient Response Specifications

Specification	Symbol	Definition & Formula
Rise Time	t_r	Time for response to rise from 10% to 90% of final value (underdamped); or 0% to 100% (overdamped). $t_r \approx \frac{1.8}{\omega_n}$
Peak Time	t_p	Time to reach the first (maximum) peak of overshoot. $t_p = \frac{\pi}{\omega_d}$
Peak Overshoot	M_p	Maximum overshoot above final value, expressed as %. $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$
Settling Time	t_s	Time for response to stay within $\pm 2\%$ (or $\pm 5\%$) of final value. $t_s \approx \frac{4}{\zeta\omega_n}$ (2% criterion)
Delay Time	t_d	Time for response to reach 50% of final value for the first time. $t_d \approx \frac{1 + 0.7\zeta}{\omega_n}$

Key Relations — Underdamped 2nd Order

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

Damped natural frequency

$$t_p = \pi / \omega_d$$

Peak time

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$$

% Peak overshoot

$$t_s \approx 4/(\zeta\omega_n)$$

Settling time (2% band)

$$t_s \approx 3/(\zeta\omega_n)$$

Settling time (5% band)

$$\zeta = \cos \theta$$

 θ = angle of pole from $-\sigma$ axis

Exam Tips — Time Domain Specs

- **Higher** $\omega_n \rightarrow$ faster response (smaller t_r, t_p, t_s).
- **Higher** $\zeta \rightarrow$ less overshoot, more damping, slower rise.
- $\zeta = 0.707$ gives $\approx 4.3\%$ overshoot — often used as optimal damping.
- M_p **depends only on** ζ ; t_p depends on ω_d ; t_s depends on $\zeta\omega_n$.
- For a 2nd order system, poles at $s = -\zeta\omega_n \pm j\omega_d$ — draw on s -plane for viva.

3. Steady-State Error and Error Constants

Steady-State Error: The difference between the desired reference and the actual output as $t \rightarrow \infty$:

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)H(s)}$$

Valid only when the closed-loop system is **stable** (Final Value Theorem condition).

3.1 Error Constants

Static Error Constants

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) \quad (\text{Position — for step input})$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) \quad (\text{Velocity — for ramp input})$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) \quad (\text{Acceleration — for parabolic input})$$

3.2 SSE vs System Type

Input	Type 0	Type 1	Type 2
Step $\left(\frac{1}{s}\right)$	$\frac{1}{1 + K_p}$	0	0
Ramp $\left(\frac{1}{s^2}\right)$	∞	$\frac{1}{K_v}$	0
Parabola $\left(\frac{1}{s^3}\right)$	∞	∞	$\frac{1}{K_a}$

Exam Tips — SSE

- **System type** = number of open-loop poles at $s = 0$ (pure integrators).
- Increasing gain K reduces SSE but may reduce stability margin.
- Adding an integrator (increasing type) eliminates SSE for lower-order inputs.
- **FVT applies only if** $sE(s)$ has all poles in LHP — check stability first.
- For non-unity feedback: $e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)}$.

4. Root Locus Method

The **Root Locus** is a graphical method that shows how the closed-loop poles move in the s -plane as the open-loop gain K varies from 0 to ∞ . It is the most powerful tool for **controller design** and **stability analysis**.

Definition: The root locus is the locus (path) of the closed-loop poles of the system $T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$ as K varies from 0 to $+\infty$.

Characteristic equation: $1 + KG(s)H(s) = 0 \Rightarrow G(s)H(s) = -\frac{1}{K}$

Angle condition: $\angle G(s)H(s) = (2q + 1) \cdot 180^\circ, q = 0, \pm 1, \pm 2, \dots$

Magnitude condition: $|KG(s)H(s)| = 1$

4.1 Rules for Constructing Root Locus

Let $G(s)H(s) = \frac{K \prod (s + z_i)}{\prod (s + p_j)}$ have m zeros and n poles ($n \geq m$).

#	Rule	Statement
1	Number of branches	Equal to n (number of open-loop poles).
2	Start & End points	Locus starts ($K = 0$) at open-loop poles ; ends ($K \rightarrow \infty$) at open-loop zeros (finite or at ∞).
3	Symmetry	Root locus is always symmetric about the real axis .
4	Real axis segments	A point on the real axis lies on the locus if the total number of real poles and zeros to its right is odd .
5	Asymptotes	$(n - m)$ branches go to ∞ along asymptotes with angles: $\phi_k = \frac{(2k + 1) \cdot 180^\circ}{n - m}$, $k = 0, 1, \dots, (n - m - 1)$
6	Centroid of asymptotes	$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$ (intersection of asymptotes on real axis)
7	Breakaway / Break-in	Points where locus leaves/enters real axis. Found from $\frac{dK}{ds} = 0$, i.e. $\frac{d}{ds}[G(s)H(s)] = 0$.
8	$j\omega$-axis crossing	Substitute $s = j\omega$ in characteristic equation; solve for ω and K . (Or use Routh array.)
9	Angle of departure	From a complex pole p_k : $\phi_{dep} = 180^\circ - \sum \angle(p_k - z_i) + \sum \angle(p_k - p_j)$
10	Angle of arrival	At a complex zero z_k : $\phi_{arr} = 180^\circ + \sum \angle(z_k - p_j) - \sum \angle(z_k - z_i)$

4.2 Root Locus — Worked Example

Example: $G(s)H(s) = K/[s(s + 2)(s + 4)]$

Given: $n = 3$ poles at $s = 0, -2, -4$; $m = 0$ zeros.

Step 1 — Branches: 3 branches.

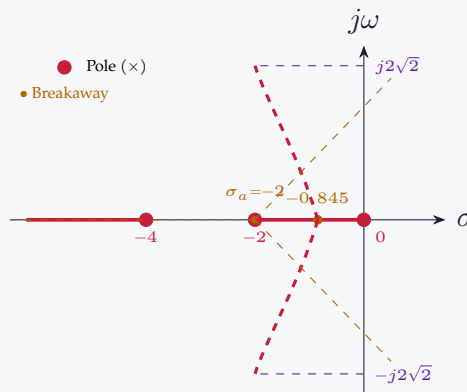
Step 2 — Real axis segments: Right of $s = 0$: 0 poles/zeros $\rightarrow$ not on locus. Between 0 and -2 : 1 pole to right $\rightarrow$ **on locus**. Between -2 and -4 : 2 poles to right $\rightarrow$ not on locus. Left of -4 : 3 poles to right $\rightarrow$ **on locus**.

Step 3 — Asymptotes: $n - m = 3$, angles = $60^\circ, 180^\circ, 300^\circ$.

Step 4 — Centroid: $\sigma_a = \frac{(0 - 2 - 4) - 0}{3} = -2$.

Step 5 — Breakaway: $\frac{dK}{ds} = 0 \Rightarrow 3s^2 + 12s + 8 = 0 \Rightarrow s \approx -0.845$ (valid, between 0 and -2).

Step 6 — $j\omega$ crossing: Characteristic eq: $s^3 + 6s^2 + 8s + K = 0$. Routh: $K_{max} = 48$, crossing at $\omega = \pm 2\sqrt{2}$.

Root Locus Sketch — $K/[s(s+2)(s+4)]$ 

Exam Tips — Root Locus

- **Locus starts at poles, ends at zeros** — memorise this first.
- **Real axis rule:** odd number of poles+zeros to the right → on locus.
- **Centroid formula:** $\sigma_a = (\sum p_i - \sum z_i)/(n - m)$ — very frequently asked.
- **Breakaway point:** solve $dK/ds = 0$ or equivalently $d[G(s)H(s)]/ds = 0$.
- **$j\omega$ crossing:** use Routh array or substitute $s = j\omega$ — gives K_{max} for stability.
- **Gain K at any point on locus:** $K = 1/|G(s)H(s)|$ at that s .

5. Compensator Design

A **compensator** (or controller) is added to the forward or feedback path to reshape the root locus so that the closed-loop poles are placed at desired locations, meeting the time-domain specifications.

Compensator: A dynamic element $G_c(s)$ inserted in the control loop to improve transient response, steady-state accuracy, or both. The two fundamental types are **Lead** (improves transient) and **Lag** (improves steady-state).

5.1 Lead Compensator

Lead Compensator: A compensator whose **zero is closer to the origin than its pole**. It adds a net *positive* phase angle to the open-loop transfer function, effectively acting like a **PD controller**. It improves **transient response** — faster rise time, reduced overshoot.

Lead Compensator Transfer Function

$$G_c(s) = K_c \frac{s + z_c}{s + p_c}, \quad p_c > z_c > 0 \quad (\text{pole farther from origin than zero})$$

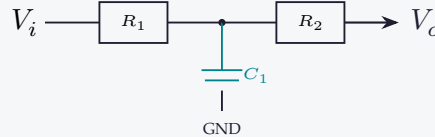
Phase contribution: $\phi = \angle(s + z_c) - \angle(s + p_c) > 0$ (positive phase lead)

Maximum phase lead: $\phi_{max} = \sin^{-1}\left(\frac{1 - \alpha}{1 + \alpha}\right)$, where $\alpha = z_c/p_c < 1$

Frequency of max lead: $\omega_{max} = \sqrt{z_c \cdot p_c}$

Op-Amp Realization of Lead Compensator

Lead Network (RC Circuit)



$$G_c(s) = \frac{R_2}{R_1 + R_2} \cdot \frac{1 + sR_1C_1}{1 + s\frac{R_1R_2}{R_1 + R_2}C_1}$$

Design Procedure (Root Locus)

Step 1: Determine desired closed-loop pole location s_d from specs (M_p, t_s, t_p).

Step 2: Calculate the **angle deficiency**: $\phi = 180^\circ - \angle G(s_d)H(s_d)$.

Step 3: Place the compensator zero z_c and pole p_c such that they contribute angle ϕ at s_d .

Step 4: Determine gain K_c from the magnitude condition: $K_c = 1/|G_c(s_d)G(s_d)H(s_d)|$.

Step 5: Verify SSE and adjust if needed.

5.2 Lag Compensator

Lag Compensator: A compensator whose **pole is closer to the origin than its zero**. It adds a net *negative* phase angle (phase lag), effectively acting like a **PI controller**. It improves **steady-state accuracy** (increases low-frequency gain) without significantly altering the transient response.

Lag Compensator Transfer Function

$$G_c(s) = K_c \frac{s + z_c}{s + p_c}, \quad 0 < p_c < z_c \quad (\text{zero farther from origin than pole})$$

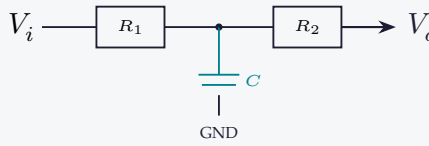
DC gain boost: $G_c(0) = K_c \frac{z_c}{p_c} = K_c \beta$, where $\beta = z_c/p_c > 1$

The lag compensator increases the **velocity constant** K_v by a factor of β , reducing SSE.

Placement rule: Place z_c and p_c very close to the origin (e.g., $z_c = 0.1, p_c = 0.01$) so they do not disturb the root locus significantly.

Op-Amp Realization of Lag Compensator

Lag Network (RC Circuit)



$$G_c(s) = \frac{1 + sR_2C}{1 + s(R_1 + R_2)C}, \quad \text{pole} < \text{zero}$$

Design Procedure (Root Locus)

- Step 1:** Design the uncompensated system to meet transient specs (find required K).
- Step 2:** Determine the factor $\beta = z_c/p_c$ needed to achieve the desired SSE improvement.
- Step 3:** Place z_c and p_c near the origin (e.g., $z_c = 0.1, p_c = z_c/\beta$).
- Step 4:** Verify that the lag compensator does not significantly shift the dominant poles.
- Step 5:** Recalculate gain K_c from magnitude condition at the desired pole.

5.3 Lead vs Lag — Comparison

Property	Lead Compensator	Lag Compensator
Pole-zero relation	$p_c > z_c$ (pole farther)	$p_c < z_c$ (pole closer)
Phase contribution	Positive (phase lead)	Negative (phase lag)
Equivalent to	PD controller	PI controller
Primary effect	Improves transient response	Improves steady-state accuracy
Bandwidth	Increases	Decreases slightly
Noise sensitivity	Higher (differentiating action)	Lower
Root locus effect	Shifts locus to the left (more stable)	Provides gain boost near origin
Typical use	Fast response needed	Low SSE needed without changing transient

5.4 Lead-Lag Compensator

Lead-Lag Compensator: Combines both lead and lag sections in cascade to simultaneously improve transient response *and* steady-state accuracy:

$$G_c(s) = K_c \frac{(s + z_1)(s + z_2)}{(s + p_1)(s + p_2)}, \quad p_1 > z_1 \text{ (lead)}, \quad p_2 < z_2 \text{ (lag)}$$

Equivalent to a **PID controller** in terms of effect.

Exam Tips — Compensators

- **Lead:** zero closer to origin than pole → phase lead → faster response.
- **Lag:** pole closer to origin than zero → phase lag → better SSE.
- **Angle deficiency** = $180^\circ - \angle G(s_d)H(s_d)$ — must be provided by lead compensator.
- **β factor** for lag = ratio z_c/p_c = SSE improvement factor.
- Lead-lag = PID equivalent; used when both transient and SSE specs must be met.
- **RC lead network:** $G_c(s) \propto (1 + sR_1C)/(1 + s\frac{R_1R_2}{R_1+R_2}C)$ — pole > zero.
- **RC lag network:** $G_c(s) \propto (1 + sR_2C)/(1 + s(R_1 + R_2)C)$ — pole < zero.

Quick Revision — Module 3

Time Response, Root Locus & Compensators | EC601

Key Formulas at a Glance

Standard 2nd order TF	$G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$
Poles	$s_{1,2} = -\zeta\omega_n \pm j\omega_d$
Damped freq.	$\omega_d = \omega_n \sqrt{1 - \zeta^2}$
Peak time	$t_p = \pi / \omega_d$
% Peak overshoot	$M_p = e^{-\pi\zeta / \sqrt{1 - \zeta^2}} \times 100\%$
Settling time (2%)	$t_s \approx 4 / (\zeta\omega_n)$
Settling time (5%)	$t_s \approx 3 / (\zeta\omega_n)$
Rise time (approx.)	$t_r \approx 1.8 / \omega_n$
SSE (step, Type 0)	$1 / (1 + K_p)$
SSE (ramp, Type 1)	$1 / K_v$
Asymptote angles	$\phi_k = (2k + 1) \cdot 180^\circ / (n - m)$
Centroid	$\sigma_a = (\sum p_i - \sum z_i) / (n - m)$
Gain at point s_d	$K = 1 / G(s_d)H(s_d) $
Lead TF	$K_c(s + z_c) / (s + p_c), p_c > z_c$
Lag TF	$K_c(s + z_c) / (s + p_c), p_c < z_c$
Max phase lead	$\phi_{max} = \sin^{-1}[(1 - \alpha) / (1 + \alpha)], \alpha = z_c / p_c$

One-Line Definitions

- ω_n : Natural frequency — speed of oscillation without damping.
- ζ : Damping ratio — controls oscillation; $\zeta = 1$ is critically damped.
- ω_d : Damped natural frequency — actual oscillation frequency in underdamped response.
- **Rise time t_r** : Time to go from 10% to 90% of final value.
- **Peak time t_p** : Time to reach first overshoot peak; $t_p = \pi / \omega_d$.
- **Overshoot M_p** : Maximum exceedance above final value; depends only on ζ .
- **Settling time t_s** : Time to stay within $\pm 2\%$ band; $t_s \approx 4 / (\zeta\omega_n)$.
- **Root locus**: Path of closed-loop poles as gain K varies $0 \rightarrow \infty$.

- **Breakaway point:** Point where locus leaves real axis; found from $dK/ds = 0$.
- **Centroid:** Intersection of asymptotes on real axis; $\sigma_a = (\sum p - \sum z)/(n - m)$.
- **Lead compensator:** Adds phase lead; improves transient; pole farther than zero.
- **Lag compensator:** Adds phase lag; improves SSE; pole closer than zero.
- **Angle deficiency:** Phase that lead compensator must supply at desired pole location.

Frequently Asked / Viva Questions

Q1. What is the standard second-order transfer function?

$$G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2).$$

Q2. What is the effect of ζ on step response?

$\zeta < 1$: underdamped (oscillatory); $\zeta = 1$: critically damped (fastest non-oscillatory); $\zeta > 1$: overdamped (sluggish).

Q3. How is M_p related to ζ ?

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% \text{ — depends only on } \zeta, \text{ not } \omega_n.$$

Q4. What is the 2% settling time formula?

$$t_s \approx 4/(\zeta\omega_n).$$

Q5. What does the root locus start and end at?

Starts at open-loop poles ($K = 0$), ends at open-loop zeros ($K \rightarrow \infty$).

Q6. State the real axis rule for root locus.

A point on the real axis is on the locus if the total number of real poles and zeros to its right is odd.

Q7. What is the centroid formula?

$$\sigma_a = (\sum \text{poles} - \sum \text{zeros}) / (n - m).$$

Q8. How do you find the breakaway point?

Differentiate $K = -1/G(s)H(s)$ with respect to s and set to zero; or solve $d[G(s)H(s)]/ds = 0$.

Q9. What is the difference between lead and lag compensator?

Lead: $p > z$, adds positive phase, improves transient. Lag: $p < z$, adds negative phase, improves SSE.

Q10. What is angle deficiency?

The phase angle that the lead compensator must contribute at the desired closed-loop pole location: $\phi = 180^\circ - \angle G(s_d)H(s_d)$.

Frequently Asked / Viva Questions (contd.)

Q11. What is the β factor in lag compensator?

$\beta = z_c/p_c > 1$; it is the factor by which the velocity constant K_v (and hence SSE improvement) is multiplied.

Q12. What is a lead-lag compensator equivalent to?

A PID controller — provides both transient improvement (lead) and SSE improvement (lag).

Damping Cases — Quick Reference

ζ value	Pole type	Response
$\zeta = 0$	$\pm j\omega_n$	Undamped — sustained oscillation
$0 < \zeta < 1$	Complex conjugate	Underdamped — decaying oscillation
$\zeta = 1$	Two equal real	Critically damped — fastest non-oscillatory
$\zeta > 1$	Two distinct real	Overdamped — no oscillation, slow

Root Locus Rules — Quick Reference

#	Rule	Formula / Statement
1	Branches	n branches (= no. of poles)
2	Start/End	$K = 0$: poles; $K \rightarrow \infty$: zeros
3	Symmetry	Symmetric about real axis
4	Real axis	Odd poles+zeros to the right
5	Asymptote angles	$(2k + 1) \cdot 180^\circ / (n - m)$
6	Centroid	$(\sum p_i - \sum z_i) / (n - m)$
7	Breakaway	$dK/ds = 0$
8	$j\omega$ crossing	$s = j\omega$ in char. eq. or Routh